(Please write your Enrollment Number) Enrollment No._____

End-Term Examination
(CBCS)
B.Tech. – IT/MAE/DMAM
(April, 2022)

Subject Code: BAS-101	Subject: Applied Mathematics-I
Time : 3 Hours	Maximum Marks : 60

Note: Q1. is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1.		**(5*4=20)**
	(a) Determine a, b, c so that the matrix A is orthogonal, where $A=\begin{vmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{vmatrix}$	
	(b) Examine the series $\sum_{n=1}^{\infty} \frac{3.6.9...3n}{4.7.10...(3n+1)} \frac{5^n}{3n+2}$ for convergence.	
	(c) Expand $e^x \cos y$ in the power of (x-1) and $(y-\frac{\pi}{4})$	
	(d) Change the order of the integration and hence evaluate $\int_0^1 \int_{e^x}^{e} \frac{dy\,dx}{logy}$	
	UNIT-1	
Q2.	**(a)** Find the eigen values and eigen vectors of the matrix $A=\begin{vmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{vmatrix}$ Also, check whether the matrix A is diagonalizable or not . **(b)** Find the rank of the matrix by reducing it to the normal form $A=\begin{vmatrix} 2 & 3 & 4 & -1 \\ 5 & 2 & 0 & -1 \\ -4 & 5 & 12 & -1 \end{vmatrix}$	**(6+4)**
Q3.	**(a)** Test for consistency and solve the following system of equations : $5x + 3y + 7z = 4$ $3x + 26y + 2z = 9$ $7x + 2y + 10z = 5$ **(b)** State Cayley Hamilton theorem and Compute A^{-1} in the powers of A for the matrix $A=\begin{vmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{vmatrix}$	**(5+5)**
	UNIT-2	
Q4.	**(a)** Test the convergence of the series whose nth term is $(n^3+1)^{\frac{1}{3}} - n$ **(b)** Test the absolute and conditional convergence of the series $\frac{1}{2^3} - \frac{1}{3^3}(1+2) + \frac{1}{4^3}(1+2+3) - \frac{1}{5^3}(1+2+3+4) + \ldots\ldots$	**(4+6)**
Q5.	**(a)** Test the convergence of the series $(\frac{2^2}{1^2} - \frac{2}{1})^{-1} + (\frac{3^3}{2^3} - \frac{3}{2})^{-2} + (\frac{4^4}{3^4} - \frac{4}{3})^{-3} + \ldots\ldots$ **(b)** Test the convergence of the series	**(5+5)**

	$x+\frac{2^2}{2!}x^2+\frac{3^3}{3!}x^3+\frac{4^4}{4!}x^4+\cdots\infty$	
	UNIT-3	
Q6.	**(a)** If $y=(x^2-1)^n$, prove that $(x^2-1)y_{n+2}+2xy_{n+1}-n(n+1)y_n=0$ **(b)** Find the minimum and maximum value of $x^3+3xy^2-15x^2-15y^2+72x$	**(5+5)**
Q7.	**(a)** Test the continuity of the function at the origin $f(x)=\begin{cases}\frac{xy}{\sqrt{x^2+y^2}}, & when\ x\neq 0, y\neq 0\\ 0, & when\ x=0, y=0\end{cases}$ **(b)** If u = f(x-y, y-z, z-x), show that $u_x+u_y+u_z=0$.	**(4+6)**
	UNIT-4	
Q8.	**(a)** Find the area bounded by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ using double integration. **(b)** Find the volume of the solid bounded by the planes $z=0, x=1, x=4, y=2, y=5$ and the surface $z=x+y$.	**(4+6)**
Q9.	**(a)** Evaluate $\int_0^1\int_0^{1-x}\int_0^{(x+y)^2} x\,dx\,dy\,dz$. **(b)** Evaluate $\iint x^2dx\,dy$ over the region in the first quadrant bounded by the rectangular hyperbola xy=16 and the lines x=y , y=0 and x=8	**(4+6)**